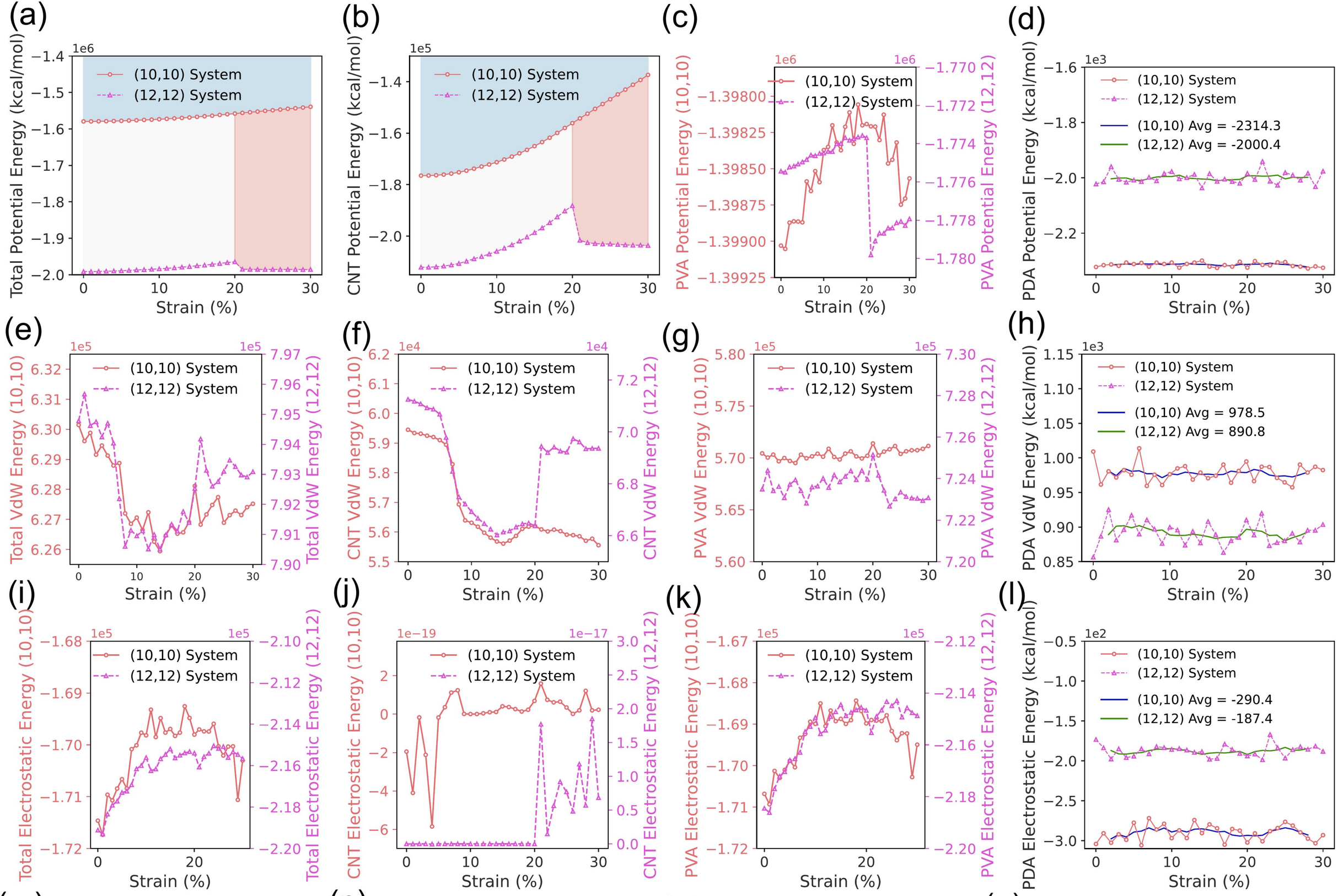
(a)
Total Potential Energy (kcal/mol)
1e6
(10,10) System
(12,12) System
Strain (%)
(b)
CNT Potential Energy (kcal/mol)
1e5
(10,10) System
(12,12) System
Strain (%)
(c)
PVA Potential Energy (10,10)
PVA Potential Energy (12,12)
1e6
1e6
(10,10) System
(12,12) System
Strain (%)
(d)
PDA Potential Energy (kcal/mol)
1e3
(10,10) System
(12,12) System
(10,10) Avg = -2314.3
(12,12) Avg = -2000.4
Strain (%)
(e)
Total VdW Energy (10,10)
Total VdW Energy (12,12)
1e5
1e5
(10,10) System
(12,12) System
Strain (%)
(f)
CNT VdW Energy (10,10)
CNT VdW Energy (12,12)
1e4
1e4
(10,10) System
(12,12) System
Strain (%)
(g)
PVA VdW Energy (10,10)
PVA VdW Energy (12,12)
1e5
1e5
(10,10) System
(12,12) System
Strain (%)
(h)
PDA VdW Energy (kcal/mol)
1e3
(10,10) System
(12,12) System
(10,10) Avg = 978.5
(12,12) Avg = 890.8
Strain (%)
(i)
Total Electrostatic Energy (10,10)
Total Electrostatic Energy (12,12)
1e5
1e5
(10,10) System
(12,12) System
Strain (%)
(j)
CNT Electrostatic Energy (10,10)
CNT Electrostatic Energy (12,12)
1e−19
1e−17
(10,10) System
(12,12) System
Strain (%)
(k)
PVA Electrostatic Energy (10,10)
PVA Electrostatic Energy (12,12)
1e5
1e5
(10,10) System
(12,12) System
Strain (%)
(l)
PDA Electrostatic Energy (kcal/mol)
1e2
(10,10) System
(12,12) System
(10,10) Avg = -290.4
(12,12) Avg = -187.4
Strain (%)
(m)
VdW Energy (kcal/mol)
1e1
(10,10) System
time (ps)
(n)
VdW Energy (kcal/mol)
1e1
(12,12) System
time (ps)
(o)
P(E)
(10,10) System
P(E)
(12,12) System
Energy (Kcal/mol)
(p)
Interphase enrgy (kcal/mol)
1e3
(10,10) System
(12,12) System
Strain (%)